

Enhancing Safety for the Visually Impaired: Visual-Language Model Based Voice Assistant for Hazard Detection and Navigation

Eesha Sutaria
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Abstract— Visual impairment is a prevalent disability worldwide that can increase the risk of serious injuries, threaten mobility, and disconnect people from their environment. Many assistive devices have been created to help visually impaired people, but none can help them with a wide variety of tasks to address both the safety and accessibility issues resulting from blindness. This work explores whether it can be effective to use a visual-language model approach to create an assistive device attachable to a white cane that serves as both a voice assistant and hazard detector for visually impaired people. The results of preliminary pilot study testing showed that the device performed effectively for nearly every category in real time, suggesting that visual language model-powered systems could potentially be used to enhance safety, accessibility, and mobility for visually impaired individuals in the future.

Keywords—*visual-language model, visually impaired, blind, hazard detection, voice assistant, accessibility, assistive device*

I. INTRODUCTION

The visually impaired population has been growing in recent years, increasing ninefold from 2012 to 2019 to reach 2.2 billion according to the World Health Organization [1]. Furthermore, around 43 million people are blind and 295 million people have moderate to severe visual impairment worldwide [2]. Thus, there is a pressing need to ensure that this community can safely and comfortably navigate their environment, even when they are in unfamiliar and outdoor places. Sighted guides are effective for this goal and oftentimes preferred for assistance by blind individuals [3]. However, blind people often feel guides take away their independence and that it can be uncomfortable and unnecessary to be constantly accompanied. Additionally, sighted guides are not always available [4]. Therefore, there have been many assistive technologies developed in the past, especially in recent years, but very few are used widely due to shortcomings and side effects such as being very limited in their practical use cases and making other tasks more difficult [5]. The most widely used device is the white cane. In a 2023 survey of 1,056 legally blind and visually impaired individuals, nearly 77% stated that they regularly use a white cane because it is affordable, long-lasting, and easy to use [6]. However, it still has limitations including not being able to detect anything above floor-level obstacles and even missing some ground hazards like water on the floor [7]. Therefore, there is a need for a new device that can help visually impaired individuals with multiple tasks to reach similar effectiveness to a sighted guide. To ensure convenience and user adoption, the device should be designed to augment the widely used white cane. This would help keep these people safe and give them the freedom and confidence to explore new environments alone.

A. Literature Review

To create an effective and beneficial device, it is imperative to be familiar with existing technology and understand the needs of the visually impaired population that must be met.

The white cane is currently the most used and hardest to beat assistive device due to its cost, reliability, and never needing a power source, while newer technologies often overwhelm the user or have a limited range of use [8]. Still the white cane cannot detect most hazards or help users feel connected to their environment. This demonstrates that there is a clear gap in technology that can adequately keep visually impaired people safe and be something they are willing to use. Skovfoged et al. [9] recommend the augmented white cane approach to assistive devices because they allow users to still use traditional orientation and mobility techniques for navigation, but also protect them from what white canes cannot detect. In order to ensure that a new product is something convenient to use and does not lose the benefit of the popular white cane, the device developed in this work should attach to a white cane and build upon it.

As explained in Hersh & Johnson [10], a book written by professors about the engineering and design of assistive devices for the visually impaired and blind, people with this disability face severe accessibility barriers in mobility due to struggles with obstacle avoidance, orientation, navigation, and access to their environment. They [10] also mentioned a barrier in information accessibility due to not being able to read text and signage. In a user study with blind participants, it was found that blind people find it valuable to be informed about their environment, specifically regarding points of interest, obstacles to avoid, and landmarks to orient themselves [11].

Additionally, without sight, head-level and fall accidents represent an important risk while walking, with thirteen percent of 300 surveyed visually impaired individuals reporting head-level accidents at least once a month and seven percent reporting falling at least once a month [12]. These numbers from Manduchi and Coughlan [12] become more alarming when taken in the context that these accidents often require medical attention, regardless of the type of mobility aid that was used when it happened. Taking this high rate of serious injury into account along with the awareness of the large mobility barriers visually impaired people face demonstrates that a solution is needed that can help users avoid obstacles to increase safety while also being able to help them overcome mobility barriers to increase accessibility to their surroundings.

To better plan which features should be a part of this product and ensure it improves from previous work, it is important to analyze technologies that have already been developed. A proposed solution outlined in Kadam et al. [13] would work as an object-detection system to warn users of dangerous objects in real time and with high accuracy. A device explained in Masud et al. [14] works as an addendum to a cane to detect objects and also notifies the user of where the object is and whether it is a hazard with sound and haptic feedback. However, both devices simply notify the user of hazardous objects around them, not any other landmark objects that could help them with navigation and orientation in their environment. According to an in-depth study with visually impaired individuals, one of the most promising types of technology to help them in their daily life is a voice-activated digital assistant [15]. Without a voice assistant, those previously discussed hazard detectors cannot provide detailed, context-aware verbal feedback or fully address the user's needs since there is no capability for users to ask questions. These devices cannot provide the assistance that a sighted guide can of taking into account what uncertainty the user may be facing, but a voice assistant may be able to take context and questions into account in providing a user experience. Including object identification and hazard detection is very important due to the mobility issues described earlier about knowing points of interest and being able to navigate places safely. Incorporating a voice assistant into that hazard detection system would be beneficial because it can help users in a way that is more personalized to what they ask for at any moment, helping them feel more connected to their environment.

A new technology that is being used increasingly for a variety of tasks is a visual-language model. A visual-language model (VLM) is an artificial intelligence model that has visual understanding, so it can take both text and image inputs and is trained to process both of these inputs while using their information together to produce an output text [16]. VLMs are currently being researched for applications in many fields, but currently there are not many VLM-based assistive devices for the visually impaired. There are no VLM-based devices that act as both a hazard detector and voice assistant for the visually impaired. However, due to their ability to take context into account, respond to questions, and take image input, they are well-suited to this project and assistive technology in general.

Thus, this work proposes a VLM-powered assistive device for the visually impaired that can be attached to a white cane through the addition of a camera, microphone, and speaker to the cane and act as a voice assistant as well as a hazard detection system. The user would be able to ask questions to an artificial intelligence-based system, which would help them orient themselves, learn about their environment, and have text read to them. Still, they would get the benefit that many assistive devices offer of being notified of hazards and objects in their path. This can increase their safety and access to mobility in a convenient way without posing major costs or introducing difficulty with other activities. It can also stop whenever the user would like and be a very easy-to-use system since everything is based on voice commands, spoken responses, and verbal warnings. In summary, this project aims to

prototype a VLM-based attachment for white canes that integrates hazard detection and voice assistant capabilities to enhance safety, navigation, and environmental awareness for visually impaired users.

II. METHODS

The goal of this research was to create an assistive device that could attach to a white cane to help visually impaired individuals stay safe and reduce the barriers to their mobility by notifying them of hazards and acting as a voice assistant. This assistant was named S.A.R.A.H., which stands for Safety Assistant for Real-time Awareness and Hazard Detection. At first, the goal was to create the physical product, but due to time constraints, it became apparent early on that this research would only be able to cover the software for this device and not the actual Raspberry Pi (a low-cost, small, high-performance computer) hardware components that would attach to a white cane.

In order to build the software, or write the code that the device would need to function, Python was the language used. Pyttsx3 engine was the text-to-speech library used and OpenCV converted the images from the ZED2 camera into a usable and interpretable format for the Gemini 2.0 Flash VLM. These libraries were chosen because they are standard and well-established for voice assistants and image analysis in Python. Python was chosen because it has these well-established libraries, which well suit this type of project. This was all run on a Windows laptop using Visual Studio Code.

A user-centered design methodology was used to develop this device. This was important because many devices have been made in the past that were helpful in theory, but in reality, were not convenient for the user because they did not build the device around the idea of making it favorable for a blind user. In fact, most of those devices were not even tested on blind people [17]. Due to limited resources, S.A.R.A.H. was also not tested on visually impaired people, but to ensure that it would be beneficial practically, the user point of view was considered in every part of the development process. Their point of view was also used to decide how the device would work and to understand what kind of information would be useful for S.A.R.A.H. to provide.

This process, as explained in the Literature Review section, led to the decision to create a voice assistant and hazard detection system. The next step was to decide the structure of the code flow to ensure that one device could do both of those tasks. The inputs would have to be camera input and voice input, since it would be a voice assistant that can see the situation and interpret the images of the environment. Many previous devices took the machine learning approach when it came to visual input, which meant that they created a database with multiple potential objects and taught the computer to recognize each object, flagging the hazardous ones [18]. However, for S.A.R.A.H., a different approach was chosen that few previous works have done. A visual-language model (VLM) was used instead, since they are already trained to be able to interpret images and allow the voice assistant to be able to answer questions about the environment beyond conventional hazards. Using a pre-existing VLM provided the flexibility to take in a much larger range of input questions from the user and be able to answer them well. Choosing which VLM to use was an important decision since this project largely depended on the model. The best VLM models that also have a free, open-source version are Gemini 2.0 Flash

(by Google), Llama (by Meta), and GPT-4 (by OpenAI). Ghosh et al. [19] conducted a thorough analysis on VLMs and analyzed the performance of each one by various benchmarks. The overall scores were 1933.4 for Gemini, 1926.6 for GPT-4, and 1542.3 for Llama, with Gemini's highest score indicating that it has the best overall performance. Other relevant information for the decision to choose Gemini is explained in Table 1.

TABLE 1. Comparison of Visual-Language Models

Model	Pros	Cons
Gemini 2.0 Flash	Fast and lightweight; designed for multimodal input; accurate visual reasoning; safety and quality prioritized for training; highest overall benchmark score	Slightly newer, with less documentation
LLaMA	High language understanding; efficient for text-only tasks; open-source	Not as well-suited for vision tasks; requires major adaptation for images; lowest overall benchmark score
GPT-4	High-quality visual-language reasoning; excellent multimodal accuracy	Dependent on cloud access; higher latency; limited control over backend; low quota for free version

For S.A.R.A.H.'s purposes, Gemini was the best choice due to the high quota of responses allowed for the free version and the high quality of visual reasoning even when given complex prompts. After deciding on a VLM, the flow of the code was planned. Taking into account the usability factor, it was important that the voice assistant capability took priority over the hazard detection. This is because when the user asks a question, it is much more convenient for them if their question is prioritized over hazard detection to ensure they get an answer quickly. Figure 1 is a flow chart that demonstrates the code flow for how the code functions.

There were several challenges in implementing this code flow. For instance, since the program uses Gemini to get a response, the whole answer comes back as one sentence, so the computer tries to speak that entire sentence at once. It would not stop in the middle of a sentence to address a question from the voice assistant. If it stopped mid-sentence, it would not be able to start speaking again. The best solution to implement the code flow was to completely restart the computer speaker (pytsx3 engine) before and after stepping in as a voice assistant as well as after each word of the hazard detection messages. However, this solution does have a limitation. To reset everything at the right times, how long it would take the text-to-speech built-in function to say each part was approximated. This was done using the rate of speech (150 words per minute) and number of words in the text. It estimates the time it will take in seconds to say a response through this formula: $\frac{\text{number of words} \times 60}{150}$. However, some words take longer or shorter amounts of time to speak, so this

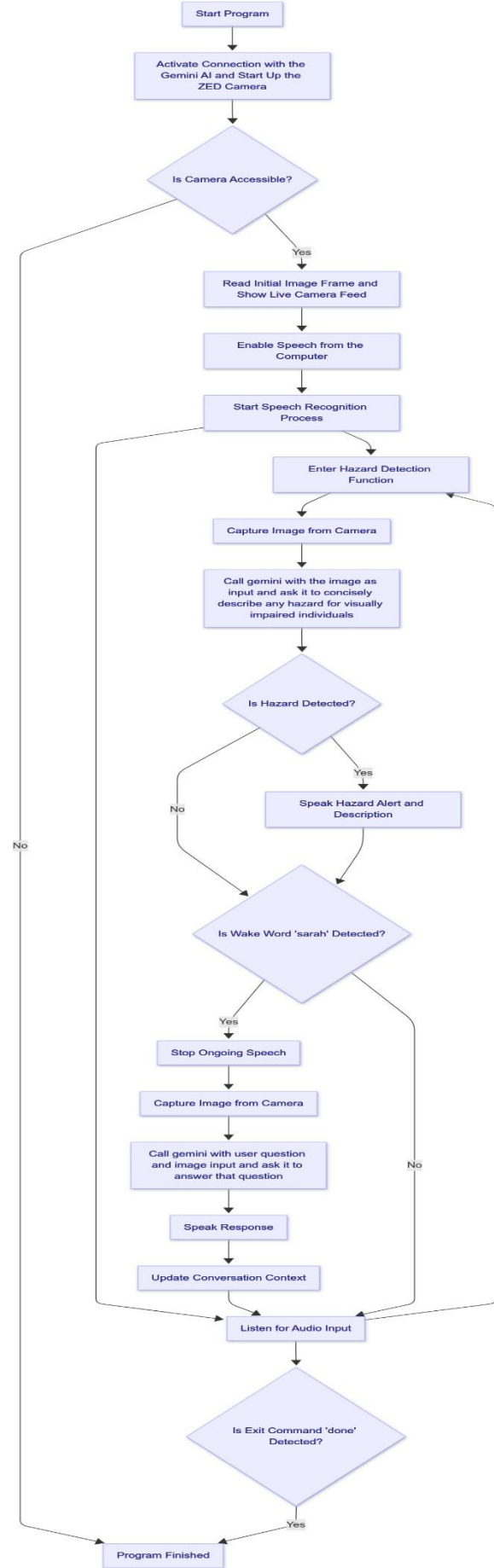


Figure 1. Code Flow Chart

doesn't always give the exact right amount of time. This can mean that sometimes the computer is waiting a little longer instead of starting back with hazard detection right away or that it cuts off the last word in a sentence of its voice assistant response. The algorithm works well to estimate time with high accuracy, which minimizes the effects of this limitation. During testing, there were no large issues with the computer waiting too long or cutting off speech.

Another important aspect of designing the product was carefully wording the prompts to the Gemini VLM and checking for precision and accuracy in Gemini's outputs during the development process. At the start, even after comparing the different visual-language models, it still wasn't completely certain that Gemini would be used until its responses were observed throughout the entire development process. Gemini's responses were very detailed and consistent among different questions and use cases. Since the software was mostly developed in the same room, the consistency of how Gemini described that room throughout testing and the development process was noted. Gemini gave nearly the same description every time with a few words switched around, and it always included all the important features of the room. This was a favorable sign and led to the pursuit of using Gemini for the assistive device. Conversation context was also added so it can answer follow-up questions appropriately.

Deciding what question to ask Gemini was difficult because there needed to be a balance between getting enough detail to keep the user safe and a concise response since too much noise can distract blind individuals from their own interpretation of their surroundings [20]. Despite the limitation of auditory information potentially being distracting, spoken output is still the best choice for this project because it will primarily be used to answer questions the user has about the environment, which means it will help the user and speak to the user only when the user wants it. For the hazards, it will only speak when it detects a hazard, which another type of warning system such as a tactile warning may not be able to sufficiently prepare users for since it would not be able to describe the issue.

Different phrasings were tested and observed for both the voice assistant prompt and the hazard detection prompt. The main features that were considered when deciding a prompt were how much irrelevant information was included and how well it took into account the user situation for each response. This evaluation was subjective and qualitative. Table 2 shows the different phrasings that were tested.

TABLE 2. *Prompts for Gemini*

Prompt Version	Voice Assistant Prompt	Hazard Detection Prompt
1	“Answer this question: {user input}.” Too vague and didn't account for the user being visually impaired	“Are there any obstacles?” Too broad; often named too much; not tailored to visual impairment
2	“Answer this question for a blind user: {user input}.” Helped focus response but the response was long	“List any visible hazards for a blind person” Listed too many objects and gave too much detail
3	“Answer this question in a brief and helpful way for someone who is blind: {user input}.” Improved conciseness, but answers were still a little long. Lacked context for answers	“Very concisely describe hazards for a blind person in this image. Head-level and floor obstacles should definitely be included. Also say where they are in relation to the user who is holding the camera. Just say 'no hazards' if you don't see any hazardous objects.” Specific and concise, consistently effective, only focused on relevant obstacles
4	“Previous context: {conversation context}. Keep your response brief and answer question for a blind person: {user input}.” Clear and concise, enough information and detail, considered context, adjusted to visually impaired people	

Figure 2 shows an image of the final product. Since only the software was developed during this project, the camera is connected to the white cane by being taped to a cardboard box and the camera connects to the computer that runs the code through a USB port. In the future, this code could be transferred to a Raspberry Pi with a camera so it can more conveniently be attached to the white cane without the user having to hold a computer.



Figure 2. *S.A.R.A.H.*

The method used to test S.A.R.A.H. was a pilot study. This method was chosen because it is widely used to test new methods. It was also used by previous works in this field that proposed new methods for assistive devices [21]. Since this device used a VLM approach and combined assistance in both the hazard detection and voice categories, the methodology was new. A pilot study is appropriate as the first way to test a new method. Furthermore, pilot studies save time and resources, which was necessary due to the time and resource constraints of this project. Thus, a pilot study was conducted as a small-scale preliminary test to help guide future development. It does have limitations, however. Since the testing is small-scale and does not include a variety of users or visually impaired individuals, the results cannot be generalized to say whether or not the device would actually help blind users. It can only provide evidence for the potential of this methodology and this device to be an effective assistive device in the future.

III. RESULTS

S.A.R.A.H. was tested using a pilot study for its ability to describe (1) Head-level obstacles, (2) Body-Level Obstacles, (3) Floor-Level Obstacles, (4) Landmarks, (5) Overall Voice Assistant Question Answering Accuracy, (6) Overall Text Reading Accuracy, (7) False Positives, and (8) False Negatives. For each of these eight categories, three users rated it on a scale of one through five and left notes on why they gave it that particular score. In the ratings out of five, one means very unacceptable, two means unacceptable, three means meets expectations, four means surpasses expectations, and five means greatly exceeds expectations. This was only tested by the developer and two other people, who were included due to the developer bias that would come from only

using the developer's own results. These results were qualitative and subjective, but this approach was chosen over quantitative testing for S.A.R.A.H. because with the sample size of three, no meaningful statistics could be performed and since this is the early stages of testing this device, subjective testing provides insights into the potential for this method in future work and demonstrates the practical usability sufficiently. S.A.R.A.H. was tested on various paths, including both indoor and outdoor environments. Specifically, it was used inside a house, on the street, in a parking lot, at a park, near mailboxes, and in a store. Each user went on one nearly identical path around a neighborhood while blindfolded to ensure that there was a control (nearly is added as a qualifier due to environmental factors that change such as animals and people coming and going). Some of the other paths were slightly varied to see how well the device could detect characteristics of a wider range of places. Each location did include multiple hazards in each category. Each user spent about an hour testing the device among the three places they tested it. However, since the results for each person were roughly the same on that shared community path, the user did not appear to make a difference on the performance of S.A.R.A.H. This consistency among users is limited by the fact that a wide variety of people did not test it. While tester results were similar across trials, the sample was still too limited to draw conclusions about user variability. A major limitation is that visually impaired individuals did not test it, but this was important for safety because the research is still fairly new and for the purposes of testing how well it can detect hazards and features of an environment, there was not a necessity to test it on visually impaired people. There were also not many people testing the product because its ability to detect hazards and describe characteristics of the environment was the feature the software was tested on, which doesn't change much for each user. Since this is a new product and approach to assistive devices, it was not also tested on its adaptability to different users on a large scale. In the future, after even more testing to ensure safety and functionality, the device should be tested by visually impaired individuals. Table 3 shows the results for the tests that were conducted on the S.A.R.A.H. device. Figure 3 demonstrates the average of the scores given by each user in each category in a chart.

TABLE 3. *Results from testing S.A.R.A.H.*

Category	Score on a scale of 1-5, 5 being best (mine, tester 1, tester 2)	Notes
Head-level hazards (description quality and detection rate)	5,5,5	Detected and described trees, signs, parking lot poles, and playground structures accurately and with enough detail to avoid them. Didn't miss any
Body-level hazards (description quality and detection rate)	5,5,5	Detected cars, mailboxes, fences, slides, and cabinets with enough detail to avoid them. Didn't miss any

Floor-level hazards (description quality and detection rate)	4,5,4	Notified the user about uneven floor tiling and sidewalks including when the floor became elevated. However, it didn't notify directly upon entering the road from the sidewalk when crossing the street.
Landmarks (description quality and detection rate)	5,5,4	Notified that there was an American flag ahead, which house numbers were on a certain street, and large signs. It did miss 1 office sign from a side view, but it got it from the head-on view when the office sign was more clearly visible.
Overall Voice Assistant Question Answering Accuracy	4,4,4	Most of the questions were answered correctly, but occasionally it asked the user follow-up questions. After that it was able to help. Some responses were a little less helpful, such as saying the park was straight and to the right because that makes it unclear when to turn right. The majority of the answers were correct and useful though.
Overall Text Reading Accuracy	5,5,5	Every sign was read correctly and a car was identified to be of a particular brand due to the logo on it. One sign had many numbers listed on it and they were all read correctly.
False Positives (reported things as hazards that weren't really hazardous)	3,4,3	There were quite a few instances where things were brought up as potential hazards, but were not truly a threat. For example, some very high palm trees were brought up as potentially hitting the user's head, but they were far too high. There were other situations like this, and they were almost all regarding head-level obstacles and items that were realistically too high up to be dangerous.
False Negatives (didn't report a potential hazard)	5,5,5	There were very, very few times it didn't report a potential hazard. The only exception was already mentioned under floor-level hazards, but that too was a rare case.

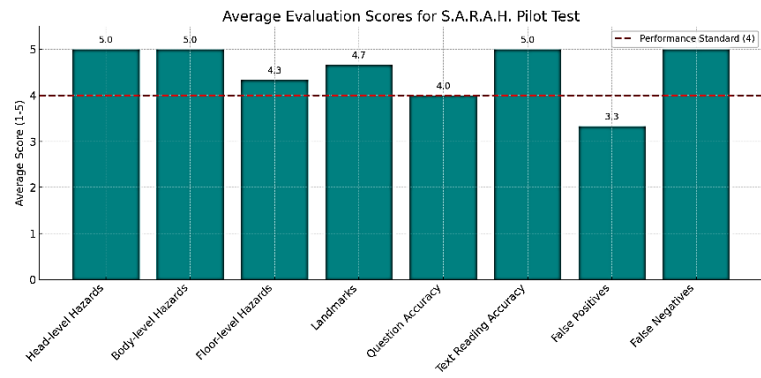


Figure 3. Average evaluation scores for S.A.R.A.H. across key performance categories. The red dashed line indicates a performance standard of 4, representing the minimum score for satisfactory functionality. Most categories exceeded this standard, suggesting strong preliminary performance in both hazard detection and voice assistance.

These results indicate that S.A.R.A.H. performed very well at detecting hazards, receiving a score of four or greater out of five in every category relating to hazard detection. A performance standard of four was used to represent a threshold of satisfactory performance. Four was chosen because these are qualitative results based on user experiences and four represents surpassing expectations. S.A.R.A.H. met or exceeded expectations in all categories except for false positives. In that category, it still scored acceptably since everyone scored it a three or above, but there should be future refinement since it was below the satisfactory performance threshold. The device detected nearly every hazard in nearly every direction on every path, with the exception of one floor hazard which was notifying the user of when the sidewalk transitioned into the road, but there were no cars coming when that occurred. In situations with cars, it did notify the user of the vehicle. The results also indicate that S.A.R.A.H. performed well as a voice assistant, notifying the user of most landmarks when asked and answering most questions correctly. The device also read all text correctly. The only area that S.A.R.A.H. did not perform very well in was false positives because the device notified the user of many things as potential hazards even though they weren't truly hazardous. While this is better than false negatives which would be more dangerous, it is still an area that the device could be improved in. It impacts blind individuals when there is a lot of audio stimuli since they mainly interact with their environment by listening to it, so these false positives should be reduced to improve usability. Overall, S.A.R.A.H. performed well suggesting that a VLM-based approach for combining a voice assistant with a hazard detector may be a worthwhile new approach to assistive devices for the visually impaired and should be further explored in the future.

IV. DISCUSSION

This device succeeded at the project goal of being a visual-language model-based approach to an assistive device for the visually impaired that can act as a voice assistant and hazard detector. Using the VLM approach allowed for the ability of a single device to conveniently work as a hazard detector and voice assistant to potentially help visually impaired individuals with two separate issues: staying safe and feeling connected to their environment. The hazard detector could help with the first part and the voice assistant

with the second. This provides preliminary evidence to promote future research into this approach towards helping the visually impaired and possibly even other disabled populations. This also meets the conditions outlined in [22] for what an assistive device should have. These guidelines include having suitable operating hours in daytime, the ability to respond in real-time, coverage of both indoor and outdoor environments, obstacle detection, a comprehensive coverage range from the ground to high up, not adding weight to the user (since it would go on the white cane in the future), robustness, and a fair cost (since all this device really needs is a camera, audio system, and computing unit).

A. Limitations

There are limitations to the methods and results of this study. Since it was not tested on a wide variety of individuals or visually impaired individuals, the results cannot determine that the device is beneficial to visually impaired people. Still, by showing that it does detect hazards and work effectively as a voice assistant, there is reason to believe that it may have potential as a beneficial and desirable visual assistant device. Additionally, since it does rely on an external artificial intelligence model, Gemini 2.0 Flash, it also faces the limitations of that model and is impacted by any issues with that model. The device also does not work without network because of this connection to Gemini. However, internet hotspots can make it easier to use this device outside of the house. S.A.R.A.H. may also not be as effective at night because the dark images may not be analyzed as accurately to find hazards. This program is also not flexible to meet the personal needs of different users. For example, it is not programmed to behave differently for fully blind individuals and those with partial blindness, which is a characteristic that would impact which hazards they may want to be notified of. The model was built with complete blindness in mind to ensure that it can help the most people. It would be beneficial to be able to meet the needs of different visually impaired groups to be more usable and helpful. Lastly, this method makes the assumption that the user is not both blind and deaf, and that they do have hearing and speaking abilities. It also assumes the user uses a white cane. This was specifically chosen as a feature to benefit most people since the majority of visually impaired individuals already use white canes and can hear. However, it could isolate some visually impaired individuals who do not use white canes or are also deaf.

B. Implications

The results of this study suggest that using visual-language models for assistive technologies could transform how devices support the visually impaired. S.A.R.A.H. demonstrated that combining hazard detection with a voice assistant into one convenient, user-focused system could potentially significantly enhance both safety and environmental awareness for visually impaired individuals. This approach opens the door to building more adaptable, multifunctional assistive devices. Additionally, the successful implementation of a visual-language model highlights a new path for future researchers who want to create assistive technology that can better understand and respond to complex real-world environments. This work provides a starting point that others can build upon, modify,

and improve based on different needs or emerging technologies. This work establishes an early framework for how visual-language models can be applied to accessibility challenges and encourages broader exploration into AI-powered assistive devices. This method could potentially help visually impaired people feel safe in unfamiliar environments without a sighted guide, increasing their independence and safety. Beyond assisting those with visual impairments, this method could be adapted to help other populations with disabilities, offering a broader impact on accessibility and mobility. Future developments in this area could lead to smarter, more affordable, and more widely adopted tools that meaningfully improve daily life for people with disabilities.

V. CONCLUSION

This study encourages future research on using visual language models to power assistive devices. It also encourages creating assistive devices with the ability to help with a wide range of issues. The resulting product of this work succeeded at the project goal of creating an assistive device to augment a white cane that could potentially enhance safety and accessibility through hazard detection and a voice assistant. In the future, this device should be converted to a Raspberry Pi so it can be attached to a cane more conveniently. The Raspberry Pi is a particularly beneficial method to use because it is cheap and can attach with a camera and audio system to transfer the full functionality of the device onto the cane [23]. Devices like the one this paper centers around should be created and tested more thoroughly, especially on visually impaired people, to further understand how it can benefit them and enhance independence. Further research could also use this visual language model intelligence approach to assist those with other disabilities and mobility issues, such as those with limited hearing or limited ability to walk.

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